Solvency Field Theory

V10.6 Companion Note

Theorem 6c: Unified Write-Rate and Flux-Burden Source Mechanism

Status: Structurally sharpened and source-wise narrowed. Formal lag/source closure remains open.

Date: April 21, 2026

Contributors: Shaun Fosmark (physical intuition and mechanism identification), Claude/Opus (translation, computation, and consistency analysis), ChatGPT/5.4 (formal verification, covariant rewrite, and theorem synthesis)

1. Theorem 6 context

Theorem 6 is the largest remaining theoretical extension in the framework. It asks for the full bridge beyond FLRW minisuperspace: a local covariant structure that unifies the FLRW count action, the strong-field exact closure, and the perturbative source sector.

The theorem naturally decomposes into three fronts:

- **Theorem 6a – observer-side bridge narrowing.** This front identifies the local completion law, narrows the bridge coefficients (α, λ), and recovers the V8.6 observer-side shell correction as the first-order sky/path realization of multiplicative forwarding through nested structures.
- **Theorem 6b – full covariant action.** This front asks for a local Lagrangian density whose FLRW reduction gives the constrained minisuperspace action, whose vacuum spherical limit gives the strong-field closure, and whose perturbative sector carries the screening corrections.
- **Theorem 6c – source-sector closure.** This front asks how radiation gravitates in SFT without introducing a preferred frame and without bolting on a separate coupling rule for massless particles.

This note records the current strongest state of **6c**. It does not claim final source-sector closure, but it sharpens the target enough that the remaining work is now formal rather than conceptual.

2. The problem

The Compton Maintenance Postulate says that every massive particle ticks at its Compton frequency and that each tick incurs one boundary write. Geometric lag is the response to that ledger cost. That story explains why rest mass gravitates.

But photons have no rest mass and no Compton frequency. If the maintenance postulate is read too narrowly, radiation would appear to cost the ledger nothing and therefore to source no gravity. That

cannot be right. Radiation shaped the early universe, drives the radiation-dominated Friedmann era, and appears in the Einstein source term through the stress-energy tensor.

So the 6c question is straightforward:

what is the universal source mechanism that makes both matter and radiation cost the ledger, while preserving the distinct ways they dilute and gravitate?

3. The unified write-rate insight

The candidate mechanism is that there is only *one* primitive operation class: a particle state is serviced by the ledger every Planck-time step. The difference between matter and radiation is not a different fee, but a different write pattern.

A massive particle at rest uses its c-motion internally. Its state is repeatedly refreshed on the same local cell at Compton rate,

$$\nu_C = \frac{mc^2}{h}.$$

This is localized, repetitive writing. It builds local lag.

A photon uses its c-motion spatially. Its state is handed off from one cell to the next as it propagates. The write address advances rather than staying fixed. Those handoff operations still cost the ledger, but the cost is transported rather than repeatedly stored at one location.

A moving massive particle interpolates between the two limits. Some of its c-motion remains internal; some is carried spatially. The split changes the spatial profile of the writes, but not the base occupancy rate.

The universal base statement is therefore

$$\text{base write rate} = \frac{E}{h}.$$

This is true for a rest particle, a moving massive particle, and a photon. The mechanism is unified at the level of information processing rather than at the level of rest mass.

4. Why energy and pressure both source gravity

The base write rate E/h captures the ledger cost of a particle *being there*. But a complete source mechanism must also account for the additional burden carried by motion through space. In GR, this is pressure or momentum flux. In SFT, it is the extra computational load associated with transporting state, not merely refreshing occupancy.

So the total source is not just occupancy load. It is

$$\text{total operation load} = \text{occupancy load} + \text{flux load}.$$

The source quantity must be written covariantly. A coordinate component such as T_{00} is not the

right final target because it risks smuggling in a preferred frame. The correct local quantities are those measured in the local clock-field frame.

Let

$$u^\mu = \frac{\nabla^\mu \Phi}{\sqrt{-\nabla_\alpha \Phi \, \nabla^\alpha \Phi}}, \qquad u^\mu u_\mu = -1,$$

be the unit timelike phase-gradient congruence from the surviving Goal 3 candidate, and let

$$h^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu$$

be the spatial projector orthogonal to u^μ.

Then the frame-projected occupancy density is

$$\rho_{\text{occ}} = T_{\mu\nu} \, u^\mu u^\nu,$$

and the frame-projected flux burden is

$$\rho_{\text{flux}} = h^{\mu\nu} T_{\mu\nu}.$$

The natural total operation density is therefore

$$\rho_{\text{ops}} = \frac{\rho_{\text{occ}} + \rho_{\text{flux}}}{h} = \frac{T_{\mu\nu} u^\mu u^\nu + h^{\mu\nu} T_{\mu\nu}}{h}.$$

The associated current is

$$J^\mu_{\text{ops}} = \rho_{\text{ops}} \, u^\mu.$$

This is the clean formal target for 6c. It preserves the original intuition, but replaces the non-covariant placeholder T_{00} with the frame-projected operation density measured by the clock field itself.

In this reading, the Einstein source is not mysterious. Energy gravitates because energy is base ledger write rate, and pressure gravitates because momentum flux is additional transport burden on the ledger.

4.1. Perfect-fluid evaluation

Take the perfect-fluid form

$$T_{\mu\nu} = (\rho + p) u_\mu u_\nu + p \, g_{\mu\nu}, \qquad u^\mu u_\mu = -1.$$

Then

$$T_{\mu\nu} u^\mu u^\nu = \rho,$$

and

$$\begin{aligned} h^{\mu\nu} T_{\mu\nu} &= (g^{\mu\nu} + u^\mu u^\nu) T_{\mu\nu} \\ &= T + \rho. \end{aligned}$$

For signature $(-,+,+,+)$,

$$T = -\rho + 3p,$$

so

$$h^{\mu\nu} T_{\mu\nu} = 3p.$$

Therefore

$$\boxed{\rho_{\text{ops}} = \frac{\rho + 3p}{h}.}$$

This is the exact source content we wanted:

$$\begin{aligned} \text{dust:} \qquad & p = 0 \Rightarrow \rho_{\text{ops}} = \frac{\rho}{h}, \\ \text{radiation:} \qquad & p = \frac{\rho}{3} \Rightarrow \rho_{\text{ops}} = \frac{2\rho}{h}. \end{aligned}$$

The factor of 2 is therefore not a bolt-on rule. It is the pressure / momentum-flux part of the same covariant operation density.

5. Relation to the existing Goal 3 source search

The unified write-rate mechanism does not replace the earlier Goal 3 work. It explains why the strongest surviving Goal 3 candidate had the form it did.

The V9 search narrowed onto the phase-gradient clock field and the active-mass scalar

$$\sigma = \frac{2}{c^2}\left(T_{\mu\nu} - \frac{1}{2} T g_{\mu\nu}\right) u^\mu u^\nu,$$

which gives ρ for dust and 2ρ for radiation. That is already a strong clue that the correct source must be built from covariant contractions of $T_{\mu\nu}$ with the clock field, not from a coordinate component or a purely formal divergence term.

The write-rate mechanism sharpens the physical meaning of that search. It says: a covariant u^μ-projected source should exist because the ledger load is the sum of frame-projected occupancy and flux burdens.

More strongly, the source content is exactly the same. Since

$$\rho_{\text{ops}} = \frac{\rho + 3p}{h},$$

while the V9 active-mass source reduces to

$$\sigma = \rho + 3p,$$

we have the direct identification

$$\boxed{h\, \rho_{\text{ops}} = \sigma.}$$

So 6c does not need a brand-new source law beyond Goal 3. The unified write-rate mechanism is the *physical interpretation* of the surviving V9 active-mass source target.

What still has to be shown is whether the lag/source coupling built from J^{μ}_{ops} reproduces, reduces to, or consistently couples to that active-mass scalar target in the full field equation.

6. Matter and radiation as one mechanism

The unified mechanism explains why matter and radiation source gravity differently without requiring separate coupling rules.

6.1. Matter

For dust-like matter, the write address stays localized. The same particles keep writing to the same local cells. In the fluid rest frame,

$$p = 0,$$

so

$$\rho^{\text{matter}}_{\text{ops}} = \frac{\rho + 3p}{h} = \frac{\rho}{h}.$$

As the universe expands, only the number density dilutes, so

$$\rho^{\text{matter}}_{\text{ops}} \propto a^{-3}.$$

This is the standard matter dilution law.

6.2. Radiation

For radiation, the write address advances. The number density still dilutes as a^{-3}, but each photon's frequency redshifts as a^{-1}. Since the base write rate per particle is $\nu = E/h$, the occupancy load per photon also falls as a^{-1}. In addition, radiation carries isotropic momentum flux, so in the fluid frame

$$p = \frac{\rho}{3}.$$

Therefore

$$\rho^{\text{rad}}_{\text{ops}} = \frac{\rho + 3p}{h} = \frac{2\rho}{h}.$$

The same mechanism therefore gives both the correct dilution law

$$\rho^{\text{rad}}_{\text{ops}} \propto a^{-4}$$

and the correct active source factor 2ρ for radiation.

So the matter/radiation split is not imposed by hand. It falls out of the same write-rate mechanism, once one distinguishes same-cell refresh from transported flux burden.

6.3. Continuous interpolation

A moving massive particle interpolates between the two limits. Part of its operation load is localized refresh; part is transport burden. The total operation load changes continuously with the balance of internal and spatial c-motion. There is therefore no discontinuity between dust, relativistic matter, and radiation — only a continuous redistribution between occupancy and flux.

7. Formal FLRW target

The first theorem-level test for 6c is FLRW radiation recovery.

In an FLRW background with homogeneous phase field $\Phi(t)$, the phase-gradient congruence becomes the comoving cosmic time direction,

$$u^\mu = (1, 0, 0, 0)$$

in comoving coordinates. Then

$$T_{\mu\nu} u^\mu u^\nu = \rho, \qquad h^{\mu\nu} T_{\mu\nu} = 3p,$$

so the operation density reduces to

$$\rho_{\text{ops}} = \frac{\rho + 3p}{h}.$$

This immediately gives the correct fluid-frame source weights:

$$\rho_{\text{ops}}^{\text{dust}} = \frac{\rho}{h}, \qquad \rho_{\text{ops}}^{\text{rad}} = \frac{2\rho}{h}.$$

It also reproduces the expected scaling behavior:

$$\rho_{\text{ops}}^{\text{matter}} \propto a^{-3}, \qquad \rho_{\text{ops}}^{\text{rad}} \propto a^{-4}.$$

The remaining formal task is not to discover the correct matter/radiation weighting — that part is now done. The remaining task is to show that a lag/source construction built from J^μ_{ops} reproduces the correct FLRW radiation gravitational coupling,

$$H^2 \propto \rho_{\text{rad}},$$

with the correct SFT normalization.

8. Bridge consistency with 6a

Theorem 6a narrowed the observer-side bridge to the screening-side relation

$$\delta F \approx +\delta\chi, \qquad \delta\chi = \frac{1}{2\pi}\,\delta\Phi,$$

with bridge pair

$$\alpha = +1, \qquad \lambda = \frac{1}{2\pi}, \qquad \alpha\lambda = \frac{1}{2\pi} \approx 0.159.$$

Any 6c source-sector closure has to be compatible with that result. The local operation source cannot re-open the sign convention problem already resolved in 6a, nor can it force an unsuppressed identity bridge. The 6a mildness bounds are therefore not just a side constraint; they are part of the consistency test for the 6c source construction.

9. What this mechanism closes

This note does not close 6c, but it does close several conceptual gaps.

1. **Why radiation gravitates.** Radiation carries energy, and energy is base ledger write rate E/h. Radiation also carries momentum flux, which adds transport burden to the ledger. Radiation therefore sources geometry through both occupancy and flux, even though it does not build the same localized same-cell lag pattern as rest mass.
2. **Why energy and pressure both gravitate.** The local source quantity is not a mysterious rule added by hand. It is the frame-projected operation density $(T_{\mu\nu}u^\mu u^\nu + h^{\mu\nu}T_{\mu\nu})/h$.
3. **Why matter and radiation dilute differently.** The a^{-3} versus a^{-4} split follows from same-cell versus next-cell writing, not from imposed Friedmann bookkeeping.
4. **Why the mass/radiation distinction is continuous.** A moving massive particle interpolates smoothly between localized and transported write patterns while keeping a unified operation-counting picture.
5. **Why the source-sector target is now sharp.** The remaining question is no longer "why should radiation gravitate at all?" It is whether a lag/source coupling built from $J^\mu_{\rm ops}$ reproduces the standard radiation gravitational response without preferred-frame pathology.

10. What remains open

Three items remain below theorem status.

10.1. Formal source-sector derivation

The candidate current

$$J^\mu_{\rm ops} = \frac{T_{\alpha\beta}u^\alpha u^\beta + h^{\alpha\beta}T_{\alpha\beta}}{h}\, u^\mu$$

still has to be inserted into the lag/source sector and shown to reproduce the correct radiation gravitational coupling.

10.2. Preferred-frame verification

Although u^μ is covariant by construction, the full perturbative source/current assembly must still be checked explicitly to show that no preferred-frame pathology reappears.

10.3. Connection to the $4\pi^2$ spine

The operation-density mechanism should ultimately be tied back to the same geometric spine that governs the rest of the framework. If every write costs one Planck area and the boundary capacity is finite, then the total operation rate divided by capacity should map onto the Hubble response through the same master-equation structure. That connection is visible conceptually, but not yet fully written in the radiation sector.

11. Bottom line

There is one underlying ledger process: state-service at rate E/h.

Rest mass concentrates that process into repetitive same-cell refresh operations. Radiation transports it through successive-cell handoffs and therefore adds momentum-flux burden. The geometry responds to the total operation load measured in the local clock-field frame,

$$\rho_{\rm ops} = \frac{T_{\mu\nu}u^\mu u^\nu + h^{\mu\nu}T_{\mu\nu}}{h} = \frac{\rho + 3p}{h}.$$

This does not yet complete Theorem 6c. But it sharpens the source mechanism enough that the remaining work is now formal rather than conceptual. The note is therefore the right kind of V10.x extension: the narrative claim is clear enough to live in the synthesis document, and the covariant source target is explicit enough to seed the later V11-level formal split.

Appendix A. Compact formal summary

Define the phase-gradient clock field

$$u^\mu = \frac{\nabla^\mu \Phi}{\sqrt{-\nabla_\alpha \Phi \nabla^\alpha \Phi}}, \qquad u^\mu u_\mu = -1.$$

Define the spatial projector

$$h^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu.$$

Define the frame-projected operation density

$$\rho_{\rm ops} = \frac{T_{\mu\nu}u^\mu u^\nu + h^{\mu\nu}T_{\mu\nu}}{h}.$$

Define the associated current

$$J^{\mu}_{\rm ops} = \rho_{\rm ops}\, u^{\mu}.$$

Then in FLRW,

$$\rho_{\rm ops} = \frac{\rho + 3p}{h},$$

with

$$\rho^{\rm dust}_{\rm ops} = \frac{\rho}{h}, \qquad \rho^{\rm rad}_{\rm ops} = \frac{2\rho}{h}.$$

Equivalently, the source content matches the surviving V9 active-mass target,

$$h\,\rho_{\rm ops} = \sigma = \rho + 3p.$$

The formal theorem target is to show that a lag/source construction built from $J^{\mu}_{\rm ops}$ reproduces the correct radiation gravitational coupling, remains compatible with the 6a bridge pair, and stays free of preferred-frame pathology.